

Master Graduation Assignment at ING

Can We Trust LLM Answers? Evidence-Based Confidence Estimation for Financial Document Analysis

Background and Motivation

Large Language Models (LLMs) have demonstrated impressive capabilities in document understanding, question answering, and information extraction. However, despite their strong performance, they remain prone to hallucinations, factual errors, and overconfidence, which limit their adoption in high-stakes environments where decisions require reliable and transparent justifications [1,2,3]. Confidence estimation, calibration, and uncertainty quantification have therefore emerged as important research directions for developing trustworthy AI systems [4].

Recent advances in Retrieval-Augmented Generation (RAG) enable LLMs to generate answers grounded in retrieved documents while providing supporting evidence and citations. Such evidence-based reasoning is particularly attractive in domains such as banking, compliance, and financial due diligence, where transparency and auditability are essential requirements. At the same time, most existing confidence estimation approaches rely primarily on model-internal signals, including verbalized confidence, self-consistency, semantic uncertainty, hidden-state representations, or activation-based uncertainty estimates [1,5,6,7,8]. Despite the effort put in this field, recent surveys highlight that confidence estimation remains an open challenge, especially when confidence scores are expected to approximate the true probability of correctness [1,2].

At ING, this challenge arises in the context of Know Your Customer (KYC), where LLMs are used to extract information from corporate and financial documents. Stakeholders reviewing AI-generated answers not only need to know what the answer is, but also how likely it is to be correct. Reliable confidence estimates could reduce manual verification effort, support risk-aware deployment of AI systems, and improve trust in AI-assisted workflows. However, relatively little work has investigated how evidence quality, evidence relevance, citation characteristics, and supporting documentation can be leveraged to estimate answer correctness in retrieval-augmented systems [9,10]. This thesis aims to investigate this research gap at the intersection of trustworthy AI, uncertainty quantification, and financial document analysis.

Approach

This project investigates whether evidence provided by retrieval-augmented LLM systems can be used to estimate the probability that an answer is correct. While most existing confidence estimation methods rely on model-internal signals such as verbalized confidence, self-consistency, uncertainty measures, or hidden-state representations, retrieval-augmented systems additionally provide citations and supporting evidence that may offer valuable information about answer reliability.

Potential research directions include evidence verification, confidence aggregation across multiple citations, probabilistic and Bayesian confidence estimation, calibration methods, and hybrid approaches that combine evidence-based and model-internal signals.

The project may use both public benchmarks and industrial datasets. Different approaches can be evaluated using established calibration and reliability metrics, such as Expected Calibration Error (ECE), Brier Score, and

reliability diagrams. The overarching goal is to determine whether evidence-based confidence estimates can better approximate the true probability of correctness and thereby improve the trustworthiness of AI-assisted decision-making in financial document analysis.

Research Question

How can evidence provided by retrieval-augmented language models be leveraged to estimate the probability that an answer is correct?

RQ1: Which properties of retrieved evidence correlate with answer correctness?

RQ2: How can multiple citations be combined?

RQ3: How well calibrated are evidence-based confidence scores?

RQ4: How do they compare with existing techniques?

Student Profile

We are looking for a motivated MSc student with:

- Solid Python programming skills.
- Has experience with data analysis using tools such as Pandas, NumPy, and Scikit-learn.
- Has completed courses in Machine Learning, Statistics, Data Mining, Artificial Intelligence, or a related field.
- Interest in applying AI research to real-world business problems; willingness to collaborate with researchers and practitioners from ING.

Planning

Weeks 1–4 Literature Review;

Weeks 5–8 Baselines;

Weeks 9–14 Approach Development;

Weeks 15–20 Evaluation;

Weeks 21–24 Analysis;

Weeks 25–30 Writing.

Practical Information

Organization: University of Twente, in collaboration with ING

Working arrangement: Hybrid, with approximately one day per week at ING’s Amsterdam office.

Internship: The student will join ING as an MSc thesis intern and receive an **internship allowance**.

Duration: Approximately 30 weeks.

Start date: Preferably as soon as possible, with flexibility depending on the student’s availability.

Supervision: Joint supervision by researchers from ING and the University of Twente.

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References

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- [3] Augenstein, I., Baldwin, T., Cha, M., et al. (2024). Factuality Challenges in the Era of Large Language Models. *Nature Machine Intelligence*.
- [4] He, W., Jiang, Z., Xiao, T., Xu, Z., & Li, Y. (2026). A Survey on Uncertainty Quantification Methods for Deep Learning. *ACM Computing Surveys*.
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- [6] Qiu, X., & Mikkilainen, R. (2024). Semantic Density: Uncertainty Quantification for Large Language Models through Confidence Measurement in Semantic Space. *NeurIPS 2024*.
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